AS · Unit 1 · WCH11 Structure, Bonding and Introduction to Organic Chemistry

Introductory organic chemistry and alkanes

Organic chemistry is the chemistry of carbon compounds, and there are millions of them. This topic gives you the tools to handle them all: how to draw and name organic molecules, how isomers arise, and how chemists describe the way bonds break. Then it puts those tools to work on the simplest family, the alkanes: where they come from, why they make good fuels, what pollution they cause, and how they react with halogens by a free-radical chain reaction.

SPEC 4.1 Hazard and risk

WHAT THIS SPEC POINT ASKS

Know the difference between a hazard and a risk.

Two different ideas

- A **hazard** is something that **could** cause harm. It is a property of the substance or the equipment itself: bromine is toxic, hexane is flammable, concentrated acid is corrosive. You can't change a hazard.
- A **risk** is the **chance** (likelihood) that the hazard actually causes harm, and how bad that harm would be, **in the way you are using it**. It depends on the amount, the concentration, the conditions (heating, open bench or fume cupboard) and who is doing the work.
- Safety precautions **reduce the risk**. They never remove the hazard. Goggles don't make an acid less corrosive; they make it less likely the acid reaches your eyes.

	Hazard	Risk
What it is	the potential to cause harm	the chance of harm happening in a given use
Can you change it?	no, it belongs to the substance	yes, by changing how you work
Example	ethanol is flammable	low with a few cm^3 in a water bath; high if heated over a Bunsen flame

Hazard symbols

Bottles carry the internationally agreed hazard pictograms (red-bordered diamonds). The ones you meet most in organic chemistry:

Pictogram	Meaning	Organic examples
Flame	flammable	alkanes, alkenes, alcohols, cyclohexane
Flame over a circle	oxidising (makes other things burn more fiercely)	potassium manganate(VII), potassium dichromate(VI)
Skull and crossbones	acutely toxic	bromine, carbon monoxide, methanol
Corrosive (liquid eating a hand and a surface)	corrosive	bromine, concentrated acids
Exclamation mark	harmful / irritant	dilute bromine water, many halogenoalkanes
Health hazard (person with a star on the chest)	carcinogenic, mutagenic, long-term harm	benzene, some halogenoalkanes, dichromates
Dead tree and fish	harmful to the environment	halogenated solvents
Gas cylinder, exploding bomb	gas under pressure; explosive	gas cylinders; organic peroxides

MARK-SCHEME WORDING

- "Flammable / inflammable" for a fuel or solvent; "toxic / corrosive / poisonous / irritant" for HCl gas. The word must match the hazard: flammable for HCl **negated** the mark.
- Pick only the symbols that apply to the substance named. Ticking every symbol scores nothing.

Often asked as: MCQ on which pictogram fits a substance; "identify a hazard of X and a precaution".

COMMON MISTAKE

Confusing the **flammable** and **oxidising** symbols. Flammable = a flame on its own (the substance burns). Oxidising = a flame on top of a circle (the substance makes something else burn).

Saying a precaution "removes the hazard". It lowers the risk only.

SPEC 4.2 Hazards of organic compounds and risk assessments

WHAT THIS SPEC POINT ASKS

Know the typical hazards of organic compounds, and why a risk assessment is needed before you work with

hazardous materials.

Why organic compounds are often hazardous

- **Flammable.** Most organic compounds burn. Small molecules are **volatile**: they evaporate easily at room temperature, so their vapour can reach a flame some distance away and ignite.
- **Toxic or harmful.** Many vapours are harmful to breathe (halogenoalkanes, bromine, benzene). Some are **carcinogenic** (benzene) or cause long-term damage.
- **Irritant or corrosive.** Bromine, acids used as catalysts (phosphoric(V), sulfuric).
- **Products can be hazardous too.** Incomplete combustion makes **carbon monoxide**; burning PVC releases **hydrogen chloride** gas.

What a risk assessment does

A risk assessment is done **before** the practical. You:

1. List every substance used and made, and look up its **hazards** (for example on a safety data sheet).
2. Decide how big the **risk** is in *this* procedure (amounts, heating, gases given off, who is doing it).
3. Choose **control measures** that lower each risk (see 4.3).
4. Plan what to do if something goes wrong (spills, skin contact, fire).

Why it is needed: so you know the hazards in advance, can reduce the risk to an acceptable level, and are ready for an accident. In a school or industry, it is also a legal requirement.

MARK-SCHEME WORDING

- A two-mark "hazard and precaution" question: M1 names the hazard; M2 gives a precaution **linked to that hazard**. Jan 2020: "flammable" + "avoid naked flames / use a heating mantle / electric water bath / small amounts".
- IGNORE generic answers: fume cupboard (for a flammable liquid), gloves, tie hair back, fire extinguisher, "do not heat".
- Oct 2020: an oxidising agent must not be mixed with alkanes because it **promotes burning** and alkanes are **flammable**.

Often asked as: name a hazard and a precaution; explain why two chemicals must be stored apart; everyday safety in an unfamiliar context.

COMMON MISTAKE

"Wear goggles and a lab coat" scores nothing on its own: these are worn in every practical. Give a precaution aimed at **this** hazard (WCH11 Jan 2020 ER).

Writing that mixing an oxidiser and a fuel "starts a fire". It **makes a fire burn more fiercely** (WCH11 Oct 2020 ER).

SPEC 4.3 Reducing risk

WHAT THIS SPEC POINT ASKS

Suggest ways to reduce risk and carry out reactions safely: (i) work on a smaller scale, (ii) take precautions that suit the particular hazard, (iii) use a different method with less hazardous substances.

Three ways to cut the risk

Method	How it helps	Example
(i) **Smaller scale**	less substance → less vapour, smaller fire, smaller spill	microscale tests in a spotting tile; a few drops of bromine water
(ii) **Precaution matched to the hazard**	targets the way harm would happen	flammable → no naked flames, heat with a **water bath** or **electric heating mantle**; toxic gas or vapour → **fume cupboard**; corrosive → gloves and goggles
(iii) **Less hazardous alternative**	swaps a dangerous chemical for a safer one	**bromine water** instead of liquid bromine; cyclohexane instead of benzene; a plant-based fuel instead of a toxic solvent

Hazards in the Unit 1 organic practicals

Practical	Hazard	Precaution
Testing with bromine	liquid bromine: very toxic, corrosive vapour	use dilute **bromine water**, small volumes, fume cupboard, gloves
Burning or heating alkanes, alkenes, cyclohexane	flammable, volatile	no naked flames near the bottle; small amounts; heat in a water bath
Halogenation in UV light	chlorine is toxic; UV damages eyes	fume cupboard; UV-protective screen or glasses
Cracking liquid paraffin	hot glass; **suck-back** of cold water; flammable gases	remove the delivery tube from the water **before** you stop heating; no flames near collected gas
Oxidation with acidified $KMnO_4$	oxidising; acid is an irritant	gloves and goggles; keep away from flammables
Incomplete combustion	carbon monoxide is toxic	good ventilation; fume cupboard

MARK-SCHEME WORDING

Oct 2025 (carbon monoxide exposure), two of: "use a fume cupboard"; "open windows / doors (ventilation)"; "wear a respirator / oxygen mask"; "take breaks, to avoid continuous exposure".

Often asked as: suggest a safer method; suggest how to reduce the risk in a given preparation.

EXAM TIP

For "suggest" questions, think: **what is the hazard → how would it hurt someone → what stops that route.** Vapour that ignites → keep flames away. Gas that poisons → fume cupboard. Liquid that burns skin → gloves.

SPEC 4.4 Homologous series and functional groups

WHAT THIS SPEC POINT ASKS

Understand what a homologous series is and what a functional group is.

Definitions

- A **functional group** is the atom or group of atoms that gives a compound its characteristic chemical reactions. Examples: C=C in alkenes, –OH in alcohols, –Br in bromoalkanes.
- A **homologous series** is a family of organic compounds that:
  - have the **same functional group**;
  - have the **same general formula**;
  - differ from the next member by **CH_2**;
  - have **similar chemical properties** (they react in the same way);
  - show a **gradual trend in physical properties** (for example, boiling temperature rises as the chain gets longer).

Series you meet in Unit 1

Series	Functional group	General formula	Example	Name ending
Alkanes	none (C–C and C–H only)	C_nH_{2n+2}	$CH_3CH_2CH_3$ propane	-ane
Cycloalkanes	ring of C–C bonds	C_nH_{2n}	cyclohexane	cyclo…ane

Series	Functional group	General formula	Example	Name ending
Alkenes	C=C	C_nH_{2n}	$CH_2=CHCH_3$ propene	-ene
Halogenoalkanes	C–X (X = F, Cl, Br, I)	$C_nH_{2n+1}X$	CH_3CH_2Br bromoethane	prefix chloro-, bromo- …
Alcohols	–OH	$C_nH_{2n+1}OH$	CH_3CH_2OH ethanol	-ol

Trend in physical properties

Alkane	Formula	Boiling temperature / °C	State at 20 °C
methane	CH_4	−162	gas
ethane	C_2H_6	−89	gas
propane	C_3H_8	−42	gas
butane	C_4H_{10}	−1	gas
pentane	C_5H_{12}	36	liquid
hexane	C_6H_{14}	69	liquid

Each extra CH_2 adds more electrons, so the London forces between molecules get stronger and more energy is needed to separate them. (Intermolecular forces are covered fully in Unit 2.)

MARK-SCHEME WORDING

Jan 2025 (same homologous series), two from: "same functional group"; "similar chemical properties"; "same general formula"; "differ by CH_2".

Often asked as: state two features of a homologous series; MCQ on which pair is in the same series; deduce a general formula for an unfamiliar series.

COMMON MISTAKE

Saying members of a series have the "same physical properties". Their physical properties **change gradually**; it is the **chemical** properties that are similar.

SPEC 4.5 IUPAC naming and drawing organic compounds

WHAT THIS SPEC POINT ASKS

(i) Name the compounds in this specification using IUPAC rules. (ii) Draw them as structural, displayed and skeletal formulae. Know the stems for C1 to C10.

Stems

Carbons	1	2	3	4	5	6	7	8	9	10
Stem	meth	eth	prop	but	pent	hex	hept	oct	non	dec

The same stems give the side-chain (alkyl) names: methyl $–CH_3$, ethyl $–C_2H_5$, propyl $–C_3H_7$.

Naming a branched alkane: the routine

1. Find the **longest continuous carbon chain**. It may bend round corners on the page. This gives the stem.
2. Number the chain from the end that gives the side chains the **lowest numbers** (lowest locants).
3. Name each side chain and give its number. Two or more of the same group: **di-, tri-, tetra-**, and **every** group needs its own number (2,2-dimethyl, not 2-dimethyl).
4. Different groups go in **alphabetical order** (ethyl before methyl; bromo before chloro before methyl). Ignore di- and tri- when you alphabetise.
5. Punctuation: **commas between numbers**, **hyphens between numbers and letters**, and the last prefix runs straight into the stem (2-methylpentane).

WORKED EXAMPLE

Worked example: 2,2,4-trimethylpentane

Structure: $(CH_3)_3CCH_2CH(CH_3)CH_3$

1. Longest chain: 5 carbons → **pentane**.
2. Three methyl groups. Numbered from the left they sit on C2, C2 and C4. From the right they sit on C2, C4, C4.
3. Compare the first point of difference: 2,**2**,4 beats 2,**4**,4. Number from the left.
4. Three methyls → **tri**methyl, and each one gets a number.

Name: **2,2,4-trimethylpentane** (this is "iso-octane", the petrol standard).

WORKED EXAMPLE

Worked example: alphabetical order

$CH_3CH_2CH(C_2H_5)CH_2CH(CH_3)CH_3$

Longest chain: 6 carbons (hexane). Numbering from the right gives methyl at 2 and ethyl at 4 (from the left it would be 3 and 5).

Alphabetical: **e**thyl before **m**ethyl → **4-ethyl-2-methylhexane**.

Naming a branched alkane

1. Name $CH_3CH(CH_3)CH_2CH(C_2H_5)CH_3$. Draw it out first: a chain of five carbons, with a methyl on carbon 2 and an ethyl on carbon 4.
2. Look for the longest chain. Following the chain into the ethyl group gives **six** carbons, not five. So the stem is **hex** and the compound is a hexane.

   The ethyl group was never really a branch: it is part of the longest chain. A "branch" of two or more carbons on the second-to-last carbon is a warning sign that you have missed the longest chain.
3. Along this six-carbon chain there are now two methyl branches. Numbering from one end puts them on carbons 2 and 4; from the other end, on 3 and 5. Choose the lower set, 2 and 4.
4. Two methyl groups means "dimethyl", and both positions are given: **2,4-dimethylhexane**.

Naming other families

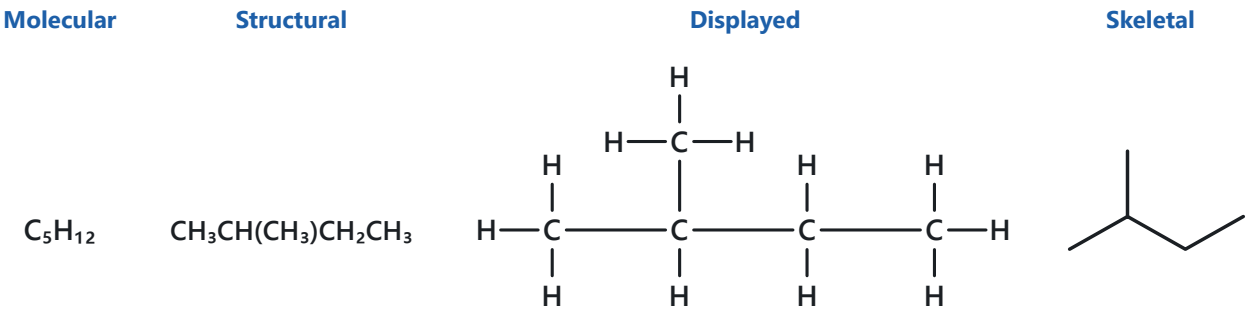
Family	Rule	Examples
Cycloalkanes	add **cyclo-** to the stem; number round the ring so substituents get the lowest numbers	cyclohexane; methylcyclopentane; 1,2-dimethylcyclobutane
Alkenes	stem + **-ene**; number the C=C from the end that gives it the **lowest number**, then place side chains	but-1-ene $CH_2=CHCH_2CH_3$; but-2-ene $CH_3CH=CHCH_3$; 2-methylbut-2-ene $(CH_3)_2C=CHCH_3$; 3-methylbut-1-ene $CH_2=CHCH(CH_3)_2$
Halogenoalkanes	prefix fluoro-, chloro-, bromo-, iodo-, alphabetical with alkyl groups	1,2-dibromopropane $CH_2BrCHBrCH_3$; 2-bromo-2-methylpropane $(CH_3)_3CBr$; 2-chloro-3-methylbutane

Family	Rule	Examples
Alcohols	stem + **-ol** with the position of the –OH	propan-1-ol $CH_3CH_2CH_2OH$; propan-2-ol $CH_3CH(OH)CH_3$
Diols	keep the "e" of the stem, add **-diol**	ethane-1,2-diol $HOCH_2CH_2OH$; propane-1,2-diol $CH_3CH(OH)CH_2OH$
Polymers	poly(monomer name)	poly(ethene); poly(propene); poly(chloroethene)

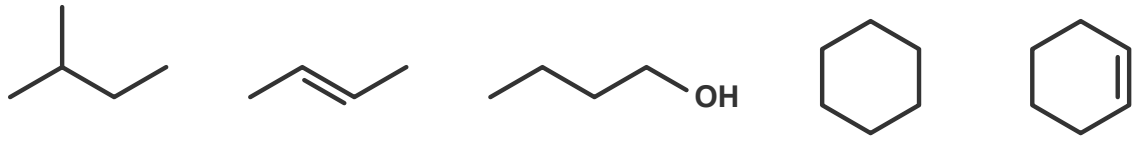
When two numbering directions tie, the group that comes first in the alphabet gets the lower number: $CH_3CHClCH(CH_3)CH_3$ is 2-chloro-3-methylbutane.

The formula types

Type	What it shows	2-methylbutane
Molecular	the actual number of each atom	C_5H_{12}
Empirical	simplest whole-number ratio	C_5H_{12} (no simpler ratio here; for ethane, C_2H_6, it is CH_3)
General	formula for a whole series	C_nH_{2n+2}
Structural	enough to identify the structure without ambiguity, group by group	$CH_3CH(CH_3)CH_2CH_3$
Displayed	**every** atom and **every** bond	see figure
Skeletal	carbon chain as zig-zag lines; each end and each corner is a carbon; H on carbon not shown; functional groups shown in full	see figure



One molecule, four formulae. In the skeletal formula, each line end and each corner is a carbon with enough hydrogens to make four bonds.



2-methylbutane | but-2-ene | butan-1-ol | cyclohexane | cyclohexene

Five skeletal formulae. In but-2-ene the second line marks the C=C double bond. Cyclohexane, C_6H_{12}, is a ring of six CH_2 groups.

Drawing tips

- **Displayed**: count four bonds on every carbon. Draw the O–H bond in an alcohol (C–O–H), not just "OH".
- **Skeletal**: never write C or CH_3 at the line ends. Do write functional groups: –OH, –Br, and the C=C as a double line. Count the corners and ends afterwards to check the carbon total.
- **Structural**: brackets show a side chain on the carbon just before it: $CH_3CH(CH_3)CH_3$.

MARK-SCHEME WORDING

- Skeletal formula: "IGNORE bond angles and lengths"; a skeletal formula with **CH_3 groups written in** is not credited (Jan 2019). The O–H in a skeletal alcohol must be bonded through the O: "do not allow O–H–C".
- If you give more than one type of formula for the same molecule, **all must be correct**.
- Names need every part: E/Z where relevant, the position of the double bond, the stem and the ending.

Often asked as: name a branched alkane (2,2,4-trimethylpentane in Jan 2023 and Oct 2024); draw a skeletal formula from a name; name a product such as 1,2-dibromopropane.

COMMON MISTAKE

"2,2-dimethyl-4-methylpentane": combine identical groups and number them all → 2,2,4-trimethylpentane (WCH11 Jan 2023 ER).

Counting the longest chain wrongly when it bends. Trace every possible chain.

Numbering from the wrong end (2,4,4- instead of 2,2,4-).

Missing "cyclo" or "di"; using "oct" for the wrong number of carbons (WCH11 Jun 2024 ER).

Giving a molecular formula when the skeletal formula is asked for, or a structural formula when displayed is asked for (WCH11 Oct 2020 ER).

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

Names of unfamiliar molecules built from the same rules, e.g. 3-ethyl-2,3-dimethylpentane in an MCQ (WCH11 Jan 2026).

Simple inorganic names using the same prefixes: S_2F_{10} is disulfur decafluoride (WCH11 Jan 2025).

SPEC 4.6 Classifying reactions

WHAT THIS SPEC POINT ASKS

Classify a reaction as addition, substitution, oxidation, reduction or polymerisation.

The five types

Type	What happens	Unit 1 example
Addition	two molecules join to give **one** product; a C=C becomes C–C	$CH_2{=}CH_2 + Br_2 \rightarrow CH_2BrCH_2Br$
Substitution	one atom or group is **replaced** by another; two products	$CH_4 + Cl_2 \rightarrow CH_3Cl + HCl$
Oxidation	gain of oxygen or loss of hydrogen (loss of electrons)	combustion; alkene + acidified $KMnO_4$ → diol
Reduction	gain of hydrogen or loss of oxygen (gain of electrons)	$CH_2{=}CH_2 + H_2 \rightarrow CH_3CH_3$
Polymerisation	many small monomer molecules join into a long chain	$n\ CH_2{=}CH_2 \rightarrow -[CH_2{-}CH_2]-_n$

How to decide

- Count the products. **One** product from two reactants → addition. **Two** products, with an atom swapped → substitution.
- Look at what the organic molecule gains or loses. Gains O or loses H → oxidation. Gains H → reduction.
- One reaction can fit two labels. Alkene + H_2 is **addition** *and* **reduction** (also called hydrogenation).
- The mechanism name adds a word in front: **free radical** substitution (4.18); **electrophilic** addition (5.6).

MARK-SCHEME WORDING

- Exact words are needed. Jan 2020, alkene + H_2 with nickel: "addition / reduction / hydrogenation". **DO NOT AWARD "electrophilic addition"** for this reaction.
- Jun 2022: an alkene + HBr needed "**electrophilic** addition"; "addition" alone was not enough there.

Often asked as: "state the type of reaction"; classify each step in a reaction scheme.

COMMON MISTAKE

Calling a substitution an "addition" (WCH11 Oct 2020 ER). Check: did the C=C disappear, or did one atom swap for another?

Calling hydrogenation "electrophilic addition". H_2 is not an electrophile; the reaction happens on the nickel surface.

SPEC 4.7 Homolytic and heterolytic bond breaking

WHAT THIS SPEC POINT ASKS

Understand that a covalent bond can break (i) homolytically, giving free radicals, or (ii) heterolytically, giving ions.

Two ways to split a bonding pair

- **Homolytic fission**: the bond breaks **evenly**. Each atom takes **one** electron from the shared pair. You get two **free radicals**.
  - Cl–Cl → Cl• + •Cl (written 2Cl•)
  - Needs energy from **UV light** or high temperature. Typical for **non-polar** bonds such as Cl–Cl, Br–Br, C–C.
- **Heterolytic fission**: the bond breaks **unevenly**. **One** atom takes **both** electrons. You get a positive ion and a negative ion.
  - $H–Br \rightarrow H^+ + Br^-$
  - The **more electronegative** atom takes the pair. Typical for **polar** bonds.

Homolytic fission



each atom keeps one electron of the pair
→ two free radicals (Cl_2 needs UV light)

Heterolytic fission



the more electronegative atom (Br) takes
both electrons → two ions

Left: two half-headed ("fish-hook") arrows, one electron to each Cl. Right: one double-headed curly arrow, both electrons to Br.

Curly-arrow rules

Arrow	Shows the movement of	Used in
Half-headed (fish-hook), one barb	**one** electron	homolytic fission, free-radical steps
Full (double-barbed) head	a **pair** of electrons	heterolytic fission, electrophilic addition

- An arrow always **starts at a bond or a lone pair** (where the electrons are) and **ends at an atom** (or between two atoms where a new bond forms).
- Never start an arrow on an atom symbol or on a + charge.

MARK-SCHEME WORDING

- Homolytic fission: "the bond breaks and **one electron goes to each atom**". Mention electrons moving; "splits equally" alone is too vague (Jan 2024 ER).
- Initiation drawn with **single-headed (half) arrows** on the bond and **dots** on both radicals; omission of the dot is penalised once only.

Often asked as: define homolytic fission; draw the homolytic fission of a given molecule (not just Cl_2); MCQ on the products of heterolytic fission.

COMMON MISTAKE

Double-headed arrows for homolytic fission (WCH11 Jun 2019, Oct 2020, Jun 2024 ER).

Showing radicals with charges (Cl^+, Cl^-) instead of dots.

Showing homolytic fission of Cl_2 when the question asked about a different molecule (WCH11 Oct 2020 ER).

SPEC 4.8 Free radicals and electrophiles

WHAT THIS SPEC POINT ASKS

Know the definitions of "free radical" and "electrophile".

Definitions to learn word for word

- **Free radical**: a species with an **unpaired electron**. Shown with a dot on the atom that carries the unpaired electron: Cl•, CH_3•, CH_3CH_2•.
- **Electrophile**: an **electron-pair acceptor**. It is attracted to a region of **high electron density** (such as a C=C double bond) and accepts a pair of electrons to form a new covalent bond. Electrophiles are positive ions or have a δ+ atom: H^+, the Hδ+ end of HBr, the Brδ+ end of a polarised Br_2.

Why radicals and electrophiles react

- A radical's unpaired electron makes it very **reactive**: it pairs up by taking an electron from another molecule.
- An electrophile is short of electrons, so it heads for electron-rich sites. "Electrophile" = *electron-loving*.

MARK-SCHEME WORDING

- Free radical: "(a species with) a **single / unpaired electron**". ALLOW "free electron". IGNORE "free radical" repeated in the answer (Jan 2026).
- Electrophile: "electron pair acceptor". (Section B mark schemes rarely ask this outright; it appears in MCQs and mechanisms.)

Often asked as: define free radical (1 mark); identify the electrophile in a mechanism; MCQ.

COMMON MISTAKE

"A radical has a free electron" or "its own electron" can lose the mark; say **unpaired**. A Cl atom with a dot and no charge is a radical; Cl^- is an ion, not a radical.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

The opposite of an electrophile, a **nucleophile**, is an **electron-pair donor** (Br^- in step 2 of electrophilic addition, OH^- in Unit 2). Nucleophile and electrophile appear side by side as MCQ options (WCH11 Jun 2022).

SPEC 4.9 Alkanes and cycloalkanes

WHAT THIS SPEC POINT ASKS

Know the general formulae of alkanes and cycloalkanes, and that they are saturated hydrocarbons (C and H only, single bonds only).

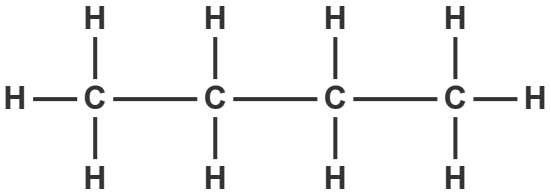
Key words

- **Hydrocarbon**: a compound of **carbon and hydrogen only**.
- **Saturated**: contains **single bonds only** (no C=C).
- **Alkane**: C_nH_{2n+2}, open chain (straight or branched). Methane CH_4, ethane C_2H_6, propane C_3H_8, butane C_4H_{10}.
- **Cycloalkane**: C_nH_{2n}, the carbons form a ring. Closing the ring removes two H atoms. Cyclohexane C_6H_{12}.

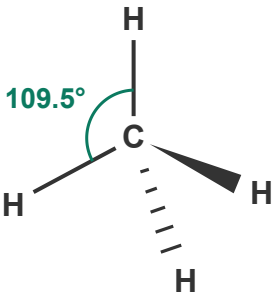
Cycloalkanes have the **same** general formula as alkenes but are still **saturated**. That's why C_6H_{12} could be hexene or cyclohexane, and only a test (5.5) tells them apart.

Structure and properties

- Each carbon makes four single bonds arranged **tetrahedrally**, bond angle **109.5°**. A chain is therefore zig-zag, not straight.
- **Unreactive** in general: C–C and C–H bonds are strong and almost **non-polar**, so electrophiles and nucleophiles are not attracted. Alkanes do burn (4.17) and react with halogens in UV light (4.18).
- **Boiling temperature** rises with chain length (more London forces). **Branching lowers** it: pentane 36 °C, 2-methylbutane 28 °C, 2,2-dimethylpropane 10 °C. Branched molecules can't pack as closely, so there is less contact between them.
- Insoluble in water; good solvents for other non-polar substances.



displayed formula of butane



methane in 3D

Left: a displayed formula shows every bond, but drawn flat it makes the molecule look square. Right: around each saturated carbon the four bonds point to the corners of a tetrahedron, 109.5° apart. A wedge comes towards you, a hashed line goes away from you, and plain lines are in the plane of the page.

MARK-SCHEME WORDING

- "Hydrocarbon: contains hydrogen and carbon **only**." "Saturated: **single (C–C) bonds only**."
- General formula questions often come as MCQs: C_nH_{2n+2} for alkanes, C_nH_{2n} for cycloalkanes, C_nH_{2n-2} for cycloalkenes (Oct 2024, Jan 2026).

Often asked as: MCQ on a general formula; "explain why cyclohexane is saturated"; "why does cyclohexane have the same formula as hexene?"

COMMON MISTAKE

"A hydrocarbon contains hydrogen and carbon" without **only**. Ethanol contains hydrogen and carbon too.

Thinking every C_nH_{2n} compound is an alkene. Cycloalkanes share the formula.

SPEC 4.10 Structural isomerism

WHAT THIS SPEC POINT ASKS

Understand structural isomerism and draw the structural isomers of a compound from its molecular formula.

Definition

Structural isomers have the **same molecular formula** but **different structural formulae** (the atoms are joined in a different order).

Three kinds

Kind	What changes	Example (same molecular formula)
Chain	the carbon skeleton (branching)	butane and 2-methylpropane (C_4H_{10})
Position	where the functional group sits	1-bromopropane and 2-bromopropane (C_3H_7Br); but-1-ene and but-2-ene (C_4H_8)
Functional group	a different functional group	propene (alkene) and cyclopropane (cycloalkane), both C_3H_6

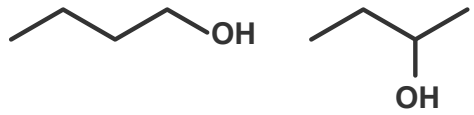
Structural isomers are **different compounds**: their physical properties differ (branched isomers boil lower), and with a different functional group their chemistry differs too.

Chain isomers, C_5H_{12}



pentane, 36 °C | 2-methylbutane, 28 °C | 2,2-dimethylpropane, 10 °C

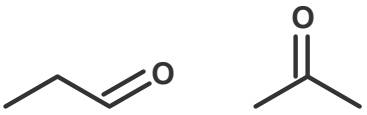
Position isomers, $C_4H_{10}O$



butan-1-ol | butan-2-ol

both $C_4H_{10}O$: –OH on a different carbon

Functional group isomers



propanal | propanone

both C_3H_6O: aldehyde and ketone

Chain isomers have different skeletons (and, as you will see below, different boiling temperatures). Butan-1-ol and butan-2-ol differ only in where the –OH is. Propanal and propanone have the same formula, C_3H_6O, but different functional groups.

How to draw all the isomers without repeats

1. Draw the **longest** chain first (all carbons in a line).
2. Shorten the chain by **one** carbon. Put that carbon on as a branch, at each **different** position. Never on an end carbon: a methyl on C1 just makes the chain longer again.
3. Shorten by **two**: add two methyls (same carbon or different carbons) or one ethyl. An ethyl branch needs to sit at least on C3 or it lengthens the chain.
4. For a functional group, move it to each distinct position on each skeleton.

5. **Name every structure.** Two drawings with the same name are the same molecule, however differently they are drawn. Also check every carbon has four bonds and the formula is right.

WORKED EXAMPLE

Worked example: isomers of $C_3H_6Br_2$

Skeleton: only one (three carbons in a line). Place two Br atoms:

- both on C1: **1,1-dibromopropane**
- on C1 and C2: **1,2-dibromopropane**
- on C1 and C3: **1,3-dibromopropane**
- both on C2: **2,2-dibromopropane**

Four isomers. "3,3-" is just 1,1- named from the wrong end.

MARK-SCHEME WORDING

- "**Same molecular formula**, different structural formula(e)" (Oct 2025). ALLOW, for alkenes, "same molecular formula but the double bond in a different position".
- Any formula type is accepted (displayed, structural, skeletal). If you give two types for one isomer, both must be correct.

Often asked as: define structural isomers (1–2 marks); draw the isomers of a given formula.

COMMON MISTAKE

Mixing up structural isomers with stereoisomers (E/Z, 5.2) or with isotopes (WCH11 Jan 2021 ER).

Drawing the same molecule twice, flipped or bent. 3-methylpentane is the one most often repeated (WCH11 Jan 2020 ER).

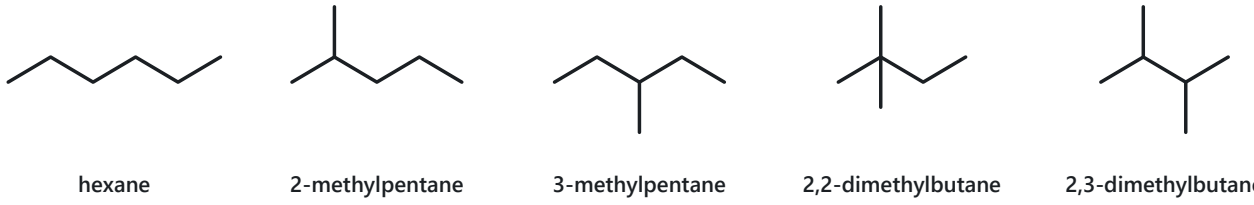
SPEC 4.11 Isomers of alkanes and cycloalkanes up to C6

WHAT THIS SPEC POINT ASKS

Draw and name the structural isomers of alkanes and cycloalkanes with up to six carbon atoms.

Alkanes

Formula	Number of isomers	Names
C_4H_{10}	2	butane; 2-methylpropane
C_5H_{12}	3	pentane; 2-methylbutane; 2,2-dimethylpropane
C_6H_{14}	5	hexane; 2-methylpentane; 3-methylpentane; 2,2-dimethylbutane; 2,3-dimethylbutane



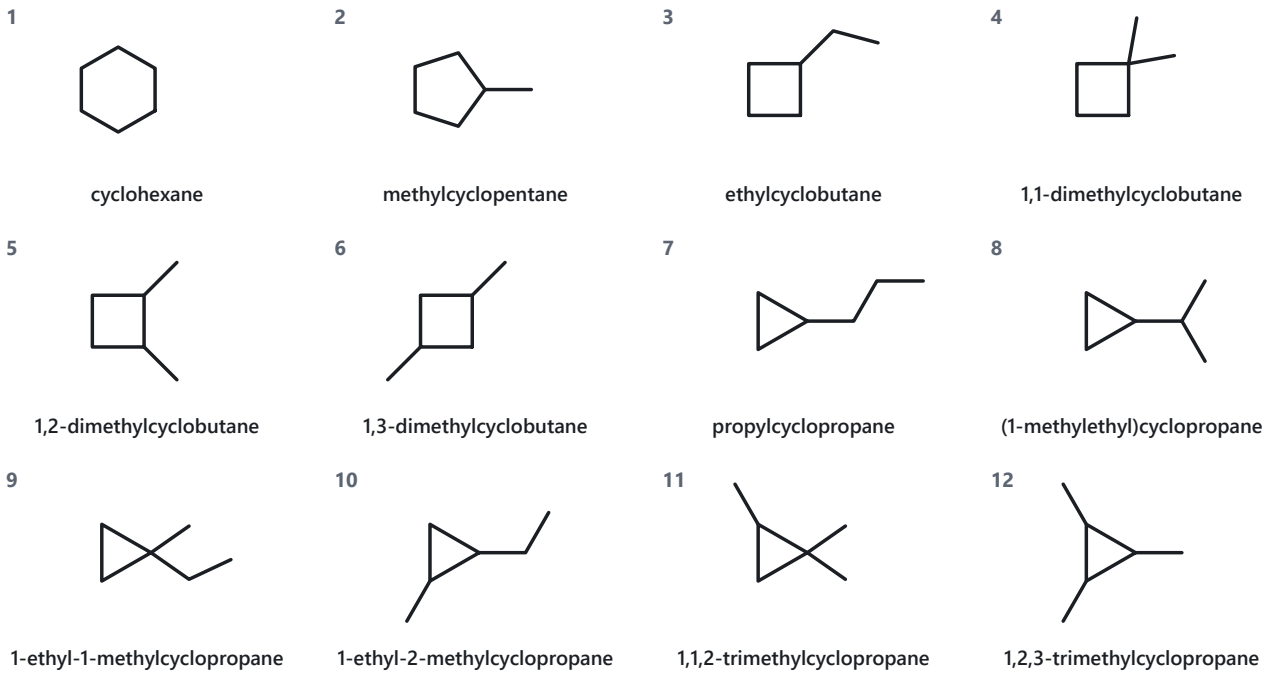
The five structural isomers of C_6H_{14} as skeletal formulae.

Check with the routine: 6 in a line (1); 5-chain + methyl on C2 or C3 (2); 4-chain + two methyls on C2,C2 or C2,C3 (2). An ethyl on a 4-chain would make a 5-chain again, so there are no more.

Cycloalkanes

Work by **ring size**, biggest first, then hang the leftover carbons on the ring.

Formula	Isomers
C_3H_6	cyclopropane
C_4H_8	cyclobutane; methylcyclopropane
C_5H_{10}	cyclopentane; methylcyclobutane; ethylcyclopropane; 1,1-dimethylcyclopropane; 1,2-dimethylcyclopropane
C_6H_{12}	12 isomers (figure below)



The twelve cycloalkane isomers of C_6H_{12}: one 6-ring, one 5-ring, four 4-rings and six 3-rings. On a ring, number so the substituents get the lowest numbers.

EXAM TIP

With a ring, "1,1-" (both groups on one carbon) is easy to forget. On a 4-ring, 1,2- (next door) and 1,3- (across) are different, but there is no 1,4-: that is 1,2- counted the other way round.

MARK-SCHEME WORDING

- Jan 2020, isomers of C_6H_{14} (other than hexane): 2-methylpentane, 3-methylpentane, 2,2-dimethylbutane, 2,3-dimethylbutane.
- Oct 2019, C_6 cycloalkanes: **DO NOT AWARD** a structure with **two or more rings** (C_6H_{12} only has room for one ring). Ignore bond angles.

Often asked as: draw the isomers of C_6H_{14} or of C_5H_{10}/C_6H_{12} with a ring; name a given isomer; MCQ: how many isomers of C_4H_9Br (Jan 2026).

COMMON MISTAKE

Including hexane when asked for isomers "of hexane". Drawing C_7H_{16} or C_6H_{12} by mistake; leaving a carbon with five bonds (WCH11 Jan 2020 ER).

Missing the cyclic isomer when the formula is C_nH_{2n}: it is the one candidates find hardest (WCH11 Jan 2023 ER). 1,1-dimethylcyclobutane was the most-missed C_6 ring isomer (WCH11 Oct 2019 ER).

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

For C_nH_{2n}, mix alkenes and cycloalkanes. Ring isomers of C_5H_{10} such as methylcyclobutane, ethylcyclopropane and 1,1-dimethylcyclopropane were asked alongside the alkenes (WCH11 Oct 2025).

C_4H_9Br has four structural isomers: 1-bromobutane, 2-bromobutane, 1-bromo-2-methylpropane, 2-bromo-2-methylpropane (WCH11 Jan 2026 MCQ).

SPEC 4.12 Alkanes as fuels: fractional distillation, cracking and reforming

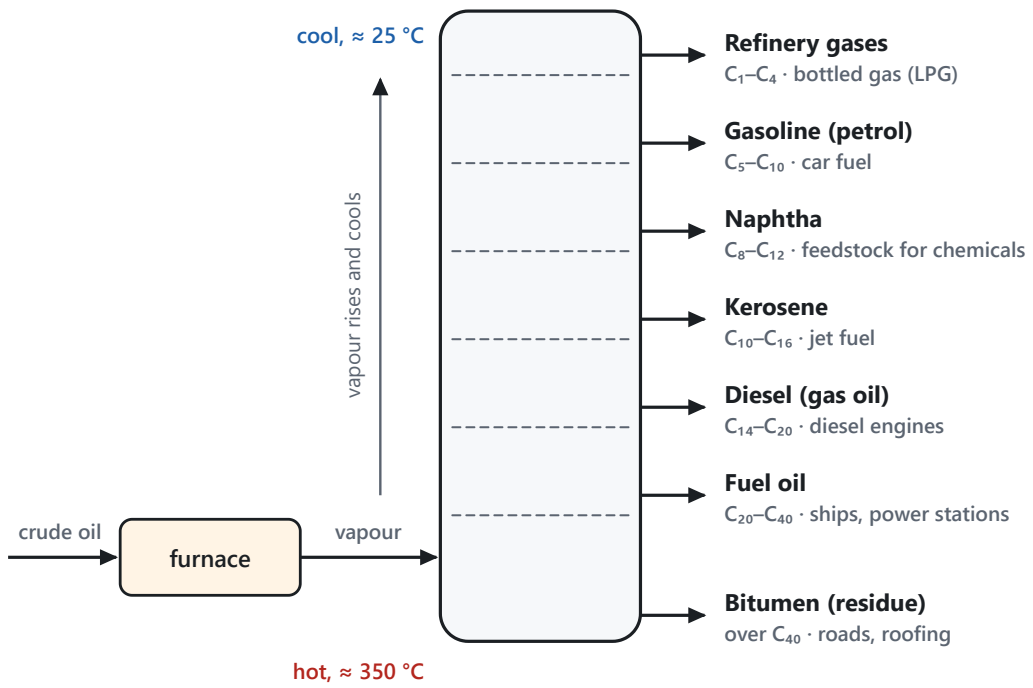
WHAT THIS SPEC POINT ASKS

Know that alkanes are used as fuels and come from crude oil by fractional distillation, cracking and reforming. Write equations for these processes.

Fractional distillation of crude oil

Crude oil is a **mixture** of hydrocarbons, mostly alkanes. Fractional distillation separates it into **fractions**: groups of compounds with similar boiling temperatures (similar chain lengths).

1. Crude oil is heated in a furnace until most of it **vaporises**.
2. The vapour enters near the bottom of a tall **fractionating column**, which is **hot at the bottom and cooler at the top** (a temperature gradient).
3. Vapour rises. Each compound **condenses** when it reaches a tray cooler than its boiling temperature, and is piped off there.
4. Longer chains have **higher boiling temperatures** (stronger London forces), so they condense **lower down**. Short chains stay as gas and leave at the top. The heaviest residue (bitumen) never vaporises.



Fractions from top to bottom: chain length and boiling temperature increase down the column. Chain-length ranges overlap and vary between refineries.

How fractional distillation separates crude oil

1. The crude oil is heated in a furnace until most of it vaporises, and the hot vapour is fed in near the bottom of the column.
2. There is a temperature gradient up the column: it is hottest at the bottom and coolest at the top. The vapours rise through trays with bubble caps and cool as they go.
3. Each hydrocarbon condenses when it reaches a tray cooler than its boiling temperature, and the liquid collected there is drawn off as a fraction.
4. Small molecules have few electrons and weak London forces, so they have low boiling temperatures and rise to the top before they condense (or leave as gas). Large molecules have strong London forces and condense near the bottom. The heaviest part never vaporises and leaves at the bottom as a residue (bitumen).

This is a physical process: only the London forces between molecules are overcome and re-formed. No covalent bonds break.

Cracking

- **What**: breaking **long-chain** alkanes into **shorter** alkanes **and alkenes**. It is thermal decomposition.
- **Why**: distillation gives too much of the long-chain fractions (fuel oil) and not enough of the short ones in demand (petrol). The alkenes made are valuable feedstock for polymers and other chemicals.
- **How**: high temperature, with or without a catalyst.
  - **Thermal cracking**: high temperature (around 450–900 °C) and high pressure; gives lots of alkenes.
  - **Catalytic cracking**: about 450 °C, low pressure, **zeolite** catalyst; gives branched alkanes and aromatic compounds for petrol.
  - In the lab: hot **aluminium oxide** or broken porcelain as the catalyst.

Equations: one alkane in, at least one shorter alkane and one alkene out. Balance carbon and hydrogen.

- $C_{10}H_{22} \rightarrow C_8H_{18} + C_2H_4$
- $C_{16}H_{34} \rightarrow C_8H_{18} + C_8H_{16}$
- $C_{12}H_{26} \rightarrow C_6H_{14} + 2C_3H_6$

Reforming

- **What**: turning **straight-chain** alkanes into **branched** alkanes, **cycloalkanes** or arenes, **keeping the same number of carbons**. Catalyst: platinum.
- **Why**: branched and cyclic hydrocarbons **burn more smoothly** in a petrol engine. They are less likely to **pre-ignite (knock)**, so the fuel has a higher **octane number**.
- Equations:
  - Isomerisation: $CH_3(CH_2)_6CH_3$ (octane) $\rightarrow$ $(CH_3)_3CCH_2CH(CH_3)CH_3$ (2,2,4-trimethylpentane), both C_8H_{18}
  - Ring formation: C_6H_{14} (hexane) $\rightarrow$ C_6H_{12} (cyclohexane) + H_2
  - C_7H_{16} (heptane) $\rightarrow$ C_7H_{14} (methylcyclohexane or cycloheptane) + H_2
  - To an arene: $C_7H_{16} \rightarrow C_6H_5CH_3$ (methylbenzene, C_7H_8) + $4H_2$

	Cracking	Reforming
Chain length	gets **shorter**	**stays the same**
Products	smaller alkane + alkene	branched or cyclic isomer (+ H_2 if a ring forms)
Purpose	meet demand for petrol; make alkenes	better petrol (less knocking)

CORE PRACTICAL

Suggested practical: cracking liquid paraffin

1. Soak mineral wool in **liquid paraffin** (a mixture of long-chain alkanes) at the bottom of a boiling tube clamped horizontally.
2. Put **aluminium oxide** (or broken porcelain) in the middle of the tube as the catalyst.
3. Heat the catalyst strongly, then flick the flame back to the paraffin so its vapour passes over the hot catalyst.
4. Collect the gas over water in test tubes. Discard the first tube (it is mostly air).
5. Shake a tube of gas with **bromine water**: orange → colourless shows an **alkene** was made.

Safety: lift the delivery tube **out of the water before you stop heating**, or cold water is sucked back into the hot tube and cracks it (suck-back). The gases are flammable.

MARK-SCHEME WORDING

- Fractional distillation (Jun 2023): "the vapour **condenses** at different levels / temperatures"; "the **higher the boiling temperature**, the **lower** the level at which it condenses". ALLOW the temperature gradient. DO NOT AWARD melting temperature; DO NOT AWARD "simple distillation".
- Why branched / cyclic: "burns more **efficiently / smoothly**", "prevents **pre-ignition / knocking / pinking**"; ALLOW "increases the octane number". IGNORE "burns more cleanly", "ignites more easily", "more volatile".
- Reforming: ALLOW "isomerisation". IGNORE "rearrangement". DO NOT AWARD "cracking".
- A reforming equation to a cycloalkane has **one** hydrocarbon product plus **H_2** (Jun 2021: $C_7H_{16} \rightarrow C_7H_{14} + H_2$).

Often asked as: name the process from an equation; write a cracking or reforming equation; explain how fractional distillation separates crude oil; explain why branched alkanes are added to petrol.

COMMON MISTAKE

Calling reforming "cracking" or "fractional distillation" (WCH11 Oct 2022 and Jan 2023 ER). If the number of carbons stays the same, it's reforming.

If the question shows a five-membered ring product, don't write cyclohexane (WCH11 Oct 2023 ER). Use the structure in the stem.

Describing the whole refinery when asked a specific point about boiling temperatures (WCH11 Oct 2024 ER).

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

Octane number and knocking are the reason for reforming (WCH11 Oct 2022).

Fractional distillation of **liquid air**: the air is first liquefied by cooling and compressing, then separated by boiling temperature. "Temperature" alone scored nothing (WCH11 Jan 2023).

SPEC 4.13 Pollutants from burning alkane fuels

WHAT THIS SPEC POINT ASKS

Know that burning alkane fuels releases carbon monoxide, oxides of nitrogen and sulfur, carbon particulates and unburned hydrocarbons.

Where each pollutant comes from

Pollutant	How it forms	Equation
Carbon monoxide, CO	incomplete combustion: **limited oxygen**	$C_8H_{18} + 8\frac{1}{2}O_2 \rightarrow 8CO + 9H_2O$
Carbon particulates (soot)	very incomplete combustion	$C_8H_{18} + 4\frac{1}{2}O_2 \rightarrow 8C + 9H_2O$
Unburned hydrocarbons	fuel that passes through the engine without burning	(none)
Oxides of nitrogen, NO and NO_2 (NO_x)	**nitrogen and oxygen from the air** react at the **high temperature** inside the engine	$N_2 + O_2 \rightarrow 2NO$; $2NO + O_2 \rightarrow 2NO_2$
Oxides of sulfur, SO_2 (and SO_3)	**sulfur impurities** in the fuel burn	$S + O_2 \rightarrow SO_2$

Complete combustion gives only CO_2 and H_2O. CO_2 is not toxic at normal levels, but it is a greenhouse gas (4.15).

MARK-SCHEME WORDING

- Pollutants (Oct 2024): nitrogen dioxide, sulfur dioxide / trioxide, carbon (particulates, soot, PM10 / PM2.5), unburnt hydrocarbons. DO NOT AWARD CO_3, SO, NO_3. IGNORE water.
- Incomplete combustion needs "**limited / insufficient** oxygen". DO NOT AWARD "no oxygen" (Jun 2021).
- NO forms "from **nitrogen and oxygen in the air** at the high temperature (of the engine)": $N_2 + O_2 \rightarrow 2NO$ (Oct 2024).

Often asked as: name a pollutant; write an equation for incomplete combustion; explain how NO forms in an engine.

COMMON MISTAKE

Saying the nitrogen comes from the fuel. Petrol contains no nitrogen; it comes from **air** (WCH11 Oct 2024 ER).

Writing a complete-combustion equation when incomplete was asked (WCH11 Jan 2023 ER).

Giving methane as the unburned hydrocarbon from a heptane fuel: the unburned hydrocarbon is the fuel itself (WCH11 Jan 2023).

SPEC 4.14 Why CO, NO_x and SO_x are a problem

WHAT THIS SPEC POINT ASKS

Understand the problems these pollutants cause, limited to why carbon monoxide is toxic and why oxides of nitrogen and sulfur are acidic.

Carbon monoxide is toxic

- CO **binds to the haemoglobin** in red blood cells (at the iron of the haem group, the site where oxygen binds).
- It binds **much more strongly** than oxygen and is hardly released (it forms carboxyhaemoglobin).
- So **less oxygen is carried** in the blood to the cells. Symptoms: headache, dizziness, drowsiness, then unconsciousness and death.
- CO is **colourless and odourless**, so you can't tell it is there.

Oxides of nitrogen and sulfur are acidic

They are **non-metal oxides**: they **dissolve in / react with water** in the atmosphere (rain, clouds) to form acids, giving **acid rain**.

- $SO_2 + H_2O \rightarrow H_2SO_3$ (sulfurous acid)
- $2SO_2 + O_2 + 2H_2O \rightarrow 2H_2SO_4$ (sulfuric acid)
- $2NO_2 + H_2O \rightarrow HNO_2 + HNO_3$ (nitrous and nitric acids)
- $4NO_2 + O_2 + 2H_2O \rightarrow 4HNO_3$

Effects of acid rain: damages limestone and marble buildings ($CaCO_3 + H_2SO_4 \rightarrow CaSO_4 + H_2O + CO_2$), corrodes metals, lowers the pH of lakes (harming fish), and damages trees and soil.

MARK-SCHEME WORDING

- CO: "toxic"; "**combines (irreversibly) with haemoglobin**" so less oxygen is carried (Oct 2021). DO NOT AWARD particulates. (In that question, if nothing else was given, "CO_2 causes suffocation" got 1 mark.)

- Acid rain (Jan 2019): oxides of nitrogen "**dissolve in / react with water**" to form "**nitric / nitrous acid(s)** / acid rain". IGNORE oxides of sulfur when nitrogen was asked about.

Often asked as: why CO is toxic (2 marks); how NO_x causes acid rain; write an equation for SO_2 or NO_2 forming an acid.

COMMON MISTAKE

Forgetting to say that the oxides **dissolve in water**. "NO_2 is acidic" is not enough (WCH11 Jan 2019 ER).

Writing about sulfur dioxide when the question named oxides of nitrogen.

Mixing up acid rain with global warming or the ozone layer. These are separate problems.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

Catalytic converters ($2CO + 2NO \rightarrow 2CO_2 + N_2$) are **not** in this unit's spec, and they are wrong as an answer to a question about removing HCl or SO_2 from waste gases (WCH11 Jun 2019 ER). Use a base instead (5.8).

SPEC 4.15 Alternative fuels: sustainability and climate change

WHAT THIS SPEC POINT ASKS

Discuss why alternative fuels are being developed: sustainability and lower emissions, including CO_2 and its link to climate change.

The case against fossil fuels

- **Finite (non-renewable).** Crude oil took millions of years to form and will run out. Using it is not **sustainable**: it doesn't meet today's needs without harming the ability of future generations to meet theirs.
- **CO_2 and climate change.** Burning fossil fuels releases carbon that was locked underground, so the CO_2 concentration in the air rises.
  - The Earth's surface absorbs sunlight and gives out **infrared** radiation.
  - **Greenhouse gases** (CO_2, CH_4, H_2O) **absorb infrared** and re-emit some of it back towards the Earth.
  - More CO_2 → **enhanced greenhouse effect** → **global warming** → climate change.
- **Other pollutants**: CO, NO_x, SO_2, particulates (4.13).

The alternatives

Fuel	Made from	For	Against
Bioethanol	fermentation of sugars from crops (sugar cane, maize)	renewable; plants absorb CO_2 as they grow	needs farmland (food vs fuel); energy for farming and distillation; lower energy per litre than petrol
Biodiesel	vegetable oils, waste cooking oil	renewable; can use waste	land use; engine changes
Biogas (methane)	anaerobic decay of waste (landfill, digesters)	uses waste; captures methane that would escape	low energy density; needs collecting
Hydrogen	electrolysis of water, or from methane	only water when burned	storage (gas, explosive); making it uses energy
Electricity (battery cars)	power stations, renewables	no exhaust emissions; quiet	only as clean as the electricity source

MARK-SCHEME WORDING

- Climate change (Jun 2019): "**carbon dioxide** (released when fossil fuels burn)" + "global warming / greenhouse effect / temperature rise". IGNORE consequences (sea level). **Acid rain and ozone** in a climate answer **negate** the mark.
- Biofuels against fossil fuels (Jun 2023): fossil fuels are **non-renewable / finite**, biofuels are **renewable**; biofuels release CO_2 that was **absorbed while the plants grew**. ALLOW "carbon neutral". DO NOT AWARD "biodegradable".
- Electric cars (Jan 2023): **state both halves**: "no CO_2 emitted (from the car)" **and** "CO_2 causes global warming"; or "can use renewable electricity". Also "no local pollution", "quieter".

Often asked as: why fossil fuels cause climate change; compare a biofuel with petrol; evaluate electric or hydrogen cars; landfill gas.

COMMON MISTAKE

Mixing acid rain, ozone and sulfur emissions into a climate-change answer (WCH11 Jun 2019 ER).

Calling biofuels "biodegradable", or claiming they release no CO_2 when burned (WCH11 Jun 2023 ER). They do release CO_2; the point is that the plants took it in first.

"Environmentally friendly", "no waste", "less pollution" with no reason. Name the gas and the effect.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

Landfill gas: waste decomposes by **anaerobic** respiration (biodegradation) to methane and CO_2, both greenhouse gases. DO NOT AWARD "thermal decomposition" or aerobic respiration (WCH11 Jun 2022).

Hexane replaced by supercritical **CO_2** as an extraction solvent: non-flammable and readily available, but needs costly high pressure (WCH11 Oct 2022).

SPEC 4.16 Carbon neutrality

WHAT THIS SPEC POINT ASKS

Apply the idea of carbon neutrality to different fuels, such as petrol, bioethanol and hydrogen.

What carbon neutral means

A fuel is **carbon neutral** if using it causes **no net change** in the amount of CO_2 in the atmosphere: the CO_2 released when it burns equals the CO_2 removed from the air to make it.

Bioethanol: neutral on paper

Stage	Equation	CO_2
Photosynthesis (crop grows)	$6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$	**6** taken in
Fermentation	$C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$	2 released
Combustion	$2C_2H_5OH + 6O_2 \rightarrow 4CO_2 + 6H_2O$	4 released

6 taken in, 2 + 4 = 6 released: in theory, no net CO_2.

In practice it is **not fully** carbon neutral, because fossil fuels supply energy for: farm machinery and fertiliser, transporting the crop and fuel, and **distilling** the ethanol out of the fermentation mixture.

Is bioethanol carbon neutral?

1. The crop takes in carbon dioxide as it grows, by photosynthesis:

$$6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$$

That is **6 CO_2 absorbed** for each glucose molecule.

2. Fermentation turns each glucose into two ethanol molecules and releases **2 CO_2**:

$$C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$$

3. Burning the two ethanol molecules releases **4 CO_2**:

$$2C_2H_5OH + 6O_2 \rightarrow 4CO_2 + 6H_2O$$

4. Total released: 2 + 4 = 6 CO_2, exactly the amount absorbed. On paper the fuel is carbon neutral.

5. In practice it is not quite: energy, mostly from fossil fuels, is used to make fertiliser, run farm machinery, transport the crop and distil the ethanol. All of that releases extra CO_2.

So biofuels reduce net CO_2 emissions compared with fossil fuels, but they are not completely carbon neutral.

Comparing three fuels

Fuel	Carbon neutral?	Why
Petrol	**no**	its carbon was locked underground for millions of years; burning it adds new CO_2 to the air and nothing takes it back out
Bioethanol	**close, not quite**	CO_2 released was absorbed by the crop recently; but energy for farming, transport and distillation usually comes from fossil fuels
Hydrogen	**depends how it is made**	$2H_2 + O_2 \rightarrow 2H_2O$, so no CO_2 when it burns. Made by electrolysis with **renewable** electricity → close to neutral; made from methane or with fossil-fuel electricity → not neutral

WORKED EXAMPLE

Worked example: energy in E10 (illustrative values)

E10 is 10% bioethanol and 90% petrol by volume. Take petrol as 34.2 MJ dm^{-3} and ethanol as 21.2 MJ dm^{-3}.

Energy from 1 dm³ of E10 = (0.90 × 34.2) + (0.10 × 21.2) = 30.78 + 2.12 = **32.9 MJ** (3 s.f.)

So you need slightly more E10 than petrol to drive the same distance. Exam questions give the values; this is a weighted mean.

MARK-SCHEME WORDING

- Carbon neutral (Jun 2019, 2 marks): the plant **takes in carbon dioxide (by photosynthesis)** as it grows **and** the **same amount** is released when the fuel burns → **no net** CO_2. "Takes in carbon" alone, or the plant taking in CO_2 by *respiration*, does not score.
- E10 (Oct 2023), three of: the ethanol is **bioethanol from plants**; made by **fermentation** of sugars; CO_2 released on combustion is **offset by CO_2 absorbed in photosynthesis**; less impact on global warming; uses less of a **finite** resource.
- Hydrogen (Oct 2021): reduces greenhouse gas emissions; fossil fuels non-renewable. DO NOT AWARD "carbon neutral" for hydrogen without discussing how it is made.

Often asked as: define carbon neutral; is fuel X carbon neutral?; advantages of E10; compare hydrogen with fossil fuels.

COMMON MISTAKE

Half the candidates scored nothing on the carbon-neutral definition (WCH11 Jun 2019 ER). Learn both halves: CO_2 **absorbed** while growing = CO_2 **released** on burning.

Answering a sustainability question about E10 with octane number or combustion efficiency from a different question (WCH11 Oct 2023 ER).

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

Energy-density calculations for blended fuels (a weighted mean) and scaling up the volume needed (WCH11 Oct 2023).

Ethene from fruit (bananas) instead of cracking: renewable, but a low yield and large amounts of crop, competing with food. "Sustainable" or "biodegradable" alone were ignored (WCH11 Jun 2019).

SPEC 4.17 Reactions of alkanes: combustion and halogenation

WHAT THIS SPEC POINT ASKS

Understand how alkanes react with (i) oxygen in air (combustion) and (ii) halogens.

Combustion

- **Complete** combustion (plenty of oxygen): CO_2 + H_2O. Very **exothermic**, which is why alkanes are fuels.
- **Incomplete** combustion (**limited** oxygen): CO and/or C (soot) + H_2O.

Fuel	Complete	Incomplete (to CO)
methane	$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$	$CH_4 + 1\frac{1}{2}O_2 \rightarrow CO + 2H_2O$
hexane	$C_6H_{14} + 9\frac{1}{2}O_2 \rightarrow 6CO_2 + 7H_2O$	$C_6H_{14} + 6\frac{1}{2}O_2 \rightarrow 6CO + 7H_2O$
heptane	$C_7H_{16} + 11O_2 \rightarrow 7CO_2 + 8H_2O$	$C_7H_{16} + 7\frac{1}{2}O_2 \rightarrow 7CO + 8H_2O$
octane	$C_8H_{18} + 12\frac{1}{2}O_2 \rightarrow 8CO_2 + 9H_2O$	$C_8H_{18} + 8\frac{1}{2}O_2 \rightarrow 8CO + 9H_2O$

WORKED EXAMPLE

Worked example: balancing a combustion equation

Complete combustion of C_7H_{16}.

1. Carbon: 7 C → **$7CO_2$**.
2. Hydrogen: 16 H → **$8H_2O$**.
3. Oxygen on the right: (7 × 2) + 8 = 22 O atoms → **$11O_2$**.

$C_7H_{16} + 11O_2 \rightarrow 7CO_2 + 8H_2O$

If the O_2 number comes out as a half (e.g. 12½ for octane), that is fine, or double everything: $2C_8H_{18} + 25O_2 \rightarrow 16CO_2 + 18H_2O$.

Balancing the complete combustion of octane

1. Write the formulae: $C_8H_{18} + O_2 \rightarrow CO_2 + H_2O$.
2. Balance carbon first: 8 C atoms make **$8CO_2$**.
3. Balance hydrogen next: 18 H atoms make **$9H_2O$**.

4 Balance oxygen last, because O_2 is an element and you can use any amount of it. On the right: 8 × 2 + 9 = 25 O atoms, which is 12½ O_2.

$$C_8H_{18}(l) + 12\tfrac{1}{2}O_2(g) \rightarrow 8CO_2(g) + 9H_2O(l)$$

Doubling everything gives whole numbers: $2C_8H_{18} + 25O_2 \rightarrow 16CO_2 + 18H_2O$. In general, C_nH_{2n+2} needs $(3n + 1)/2$ moles of O_2.

Halogenation

- Alkane + chlorine or bromine **in UV light** (or sunlight) → halogenoalkane + hydrogen halide. In the dark at room temperature there is **no reaction**.
- It is a **substitution**: a hydrogen atom is replaced by a halogen atom.
  - $CH_4 + Cl_2 \rightarrow CH_3Cl + HCl$ (chloromethane)
  - $C_2H_6 + Br_2 \rightarrow C_2H_5Br + HBr$ (bromoethane)
- With bromine you see the **orange-brown colour fade** as the bromine is used up, and steamy fumes of HBr.

MARK-SCHEME WORDING

- Combustion equations need correct formulae and balancing; a question may ask for complete **or** incomplete specifically (Jun 2021).
- Condition for halogenation: "**ultraviolet light**" (ALLOW sunlight / UV radiation; IGNORE heat).

Often asked as: write the equation for complete or incomplete combustion of a named alkane; state the condition for halogenation; overall equation for a substitution.

COMMON MISTAKE

Writing "no oxygen" for incomplete combustion. It is **limited** oxygen.

Saying alkanes react with bromine water in the dark. They don't (that is why bromine water tests for alkenes, 5.5).

SPEC 4.18 Free-radical substitution: the mechanism

WHAT THIS SPEC POINT ASKS

Understand the free-radical substitution mechanism for an alkane and a halogen: (i) radicals as species with an unpaired electron, shown by a dot; (ii) the initiation step with curly half-arrows; (iii) propagation and

termination; (iv) why the reaction is of limited use in synthesis.

The mechanism for methane + chlorine (UV light)

1. Initiation: radicals are made.

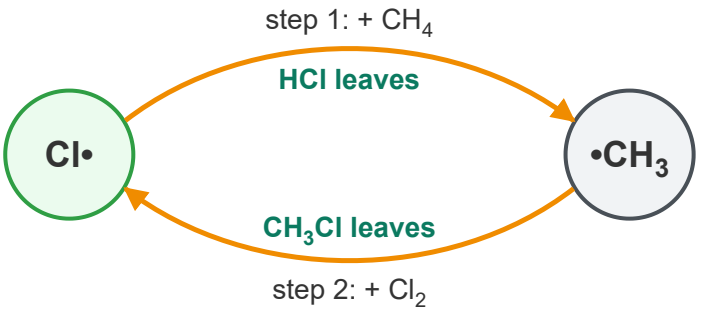
- UV light gives the energy to break the Cl–Cl bond **homolytically**. Draw **two half-headed arrows** starting from the middle of the Cl–Cl bond, one ending on each Cl.
- $Cl_2 \rightarrow 2Cl\bullet$
- The Cl–Cl bond breaks (not C–H) because it is the **weakest** bond present.



each chlorine atom keeps one electron from the shared pair

2. Propagation: a chain reaction. Each step uses up one radical and makes another.

- Step 1: $Cl\bullet + CH_4 \rightarrow \bullet CH_3 + HCl$ (the chlorine radical takes an H atom; a methyl radical is left)
- Step 2: $\bullet CH_3 + Cl_2 \rightarrow CH_3Cl + Cl\bullet$ (the methyl radical takes a Cl atom; a new chlorine radical is released)
- The Cl• made in step 2 goes back to step 1. One initiation can cause thousands of propagation cycles.
- Add the two steps: $CH_4 + Cl_2 \rightarrow CH_3Cl + HCl$ (the overall equation).



the radical is regenerated every time round, so the cycle keeps going

The two propagation steps form a loop. Each time round, one CH_4 and one Cl_2 are used up, one HCl and one CH_3Cl are made, and the chlorine radical comes back out.

3. Termination: radicals are removed. Two radicals meet and their unpaired electrons form a bond.

- $Cl\bullet + Cl\bullet \rightarrow Cl_2$

- •CH_3 + Cl• → CH_3Cl
- •CH_3 + •CH_3 → C_2H_6 (ethane)

The mechanism: methane and chlorine

1. **Initiation.** UV light provides the energy to break the Cl–Cl bond by homolytic fission, making two chlorine radicals:

$$\mathrm{Cl_2 \rightarrow 2Cl^{\bullet}}$$

Cl–Cl is the weakest bond present (242 kJ mol^{-1}, against 413 kJ mol^{-1} for C–H), so UV breaks it and leaves the C–H bonds alone.

2. **Propagation, step 1.** A chlorine radical takes a hydrogen atom from a methane molecule, forming HCl and leaving a methyl radical:

$$\mathrm{Cl^{\bullet} + CH_4 \rightarrow HCl + {^{\bullet}CH_3}}$$

3. **Propagation, step 2.** The methyl radical takes a chlorine atom from a Cl_2 molecule, forming the product and a new chlorine radical:

$$\mathrm{{^{\bullet}CH_3} + Cl_2 \rightarrow CH_3Cl + Cl^{\bullet}}$$

*Every propagation step has one radical on each side. The Cl• made in step 2 goes straight back into step 1, so one initiation can drive thousands of cycles: a **chain reaction**. Add the two propagation steps together and the radicals cancel, leaving the overall equation.*

4. **Termination.** Whenever two radicals meet, their unpaired electrons pair up to form a covalent bond, and no new radical is made. This removes radicals and stops chains:

$$\mathrm{Cl^{\bullet} + Cl^{\bullet} \rightarrow Cl_2} \qquad \mathrm{{^{\bullet}CH_3} + Cl^{\bullet} \rightarrow CH_3Cl} \qquad \mathrm{{^{\bullet}CH_3} + {^{\bullet}CH_3} \rightarrow CH_3CH_3}$$

The last step makes ethane, an impurity in the product. Write it with structural formulae (CH_3CH_3) so it is clear the two carbons have joined.

Any other alkane: same pattern

Put the radical dot on the carbon that **lost the H atom**. For ethane + chlorine:

- Initiation: Cl_2 → 2Cl•
- Propagation: Cl• + CH_3CH_3 → CH_3CH_2• + HCl

- Propagation: $CH_3CH_2\bullet + Cl_2 \rightarrow CH_3CH_2Cl + Cl\bullet$
- Termination: $CH_3CH_2\bullet + Cl\bullet \rightarrow CH_3CH_2Cl$; $2CH_3CH_2\bullet \rightarrow C_4H_{10}$ (butane); $2Cl\bullet \rightarrow Cl_2$

Two alkyl radicals joining always give a hydrocarbon with **twice** the carbons: 2 butyl radicals → octane, C_8H_{18}.

Why it has limited use in synthesis

- **Further substitution.** The product still has C–H bonds, so Cl• attacks it too: $CH_3Cl \rightarrow CH_2Cl_2 \rightarrow CHCl_3 \rightarrow CCl_4$. You get a **mixture**.
  - $Cl\bullet + CH_3Cl \rightarrow \bullet CH_2Cl + HCl$, then $\bullet CH_2Cl + Cl_2 \rightarrow CH_2Cl_2 + Cl\bullet$
- **Substitution at different positions.** Propane gives both 1-chloropropane (H removed from an end carbon) and 2-chloropropane (H removed from the middle carbon).
- **Termination** adds other products (e.g. butane from ethane).
- The mixture must be separated, and the yield of any one product is low.
- **Controlling it**: an **excess of the alkane** makes the monosubstituted product more likely (Cl• is more likely to hit an alkane than a chloroalkane). An **excess of halogen** pushes it all the way to full substitution.

WORKED EXAMPLE

Worked example: full substitution with excess bromine

Hexane reacts with a large excess of bromine in UV light. Every H is replaced, so the product is C_6Br_{14} (not $C_6H_{13}Br$).

$M_r = (6 \times 12.0) + (14 \times 79.9) = 72.0 + 1118.6 = 1190.6$

% Br by mass = 1118.6 ÷ 1190.6 × 100 = **94.0%**

MARK-SCHEME WORDING

- Name: "**(free) radical substitution**", both words. DO NOT AWARD heterolytic / nucleophilic / electrophilic. IGNORE homolytic, photochemical.
- Condition: "UV light" (ALLOW sunlight; IGNORE heat).
- Initiation: $Cl_2 \rightarrow 2Cl\bullet$ with **half-arrows**. Initiation = **one** equation, propagation = **two**, termination = **one** (e.g. a six-mark question naming each step, Jun 2021).
- Propagation, ethane (Jan 2026, 1 mark each): $Cl\bullet + CH_3CH_3 \rightarrow CH_3CH_2\bullet + HCl$; $CH_3CH_2\bullet + Cl_2 \rightarrow CH_3CH_2Cl + Cl\bullet$. ALLOW $C_2H_5\bullet$. Missing dots penalised **once only**.
- Limitation (Jan 2026): "**further substitution** (reactions) / more than one substitution product". IGNORE "further reactions", "low yield", "side reactions", "atom economy".

Often asked as: the whole mechanism for a named alkane (hexane, propane, butane, cyclohexane, 2,2,4-trimethylpentane); a termination step that makes a hydrocarbon; why the method is not a good way to make one product; a % by mass or % yield calculation.

COMMON MISTAKE

A propagation step that makes **two** radicals, or makes **H•** (WCH11 Jan 2020, Oct 2023 ER). Each propagation step has exactly one radical on each side.

Radicals drawn with + or − charges, or the dot missing (WCH11 Oct 2020, Jan 2020 ER).

Giving Cl_2 or a chloroalkane when the question asks for a termination step that makes **a hydrocarbon** (WCH11 Jan 2019 ER).

Using methane from memory when the question named hexane (WCH11 Oct 2023 ER). Use the formulae in the stem and balance each step.

"UV light breaks bonds" without saying **which** bond: it breaks the **Cl–Cl** (or Br–Br) bond homolytically (WCH11 Oct 2023 ER).

Writing more than one equation when the question asks for "a propagation step": only the first is marked (WCH11 Oct 2020 ER).

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

Mechanisms for alkanes other than methane: hexane with bromine (WCH11 Jan 2024), propane (WCH11 Jan 2026), butane (WCH11 Jun 2025), cyclohexane (WCH11 Oct 2019).

Two termination products that are hydrocarbons, drawn as displayed formulae (WCH11 Oct 2020).

With excess halogen, the fully substituted product (e.g. C_6Br_{14}) in a calculation (WCH11 Jan 2020).

Quick checks

QUICK CHECK

A teacher uses 0.1 mol dm^{-3} sodium hydroxide instead of 2 mol dm^{-3} sodium hydroxide in a class practical. What has changed?

- The hazard is lower but the risk is the same.
- Both the hazard and the risk are unchanged.
- The risk is lower; the hazard of sodium hydroxide itself is unchanged.
- The solution is no longer a hazard.

A hazard is a property of the substance, so sodium hydroxide is still corrosive or irritant. Using it more dilute reduces the chance and severity of harm, which is the risk.

QUICK CHECK

What is the empirical formula of hexane?

- C_6H_{14}
- C_3H_7
- CH_2
- C_nH_{2n+2}

Hexane is C_6H_{14}; dividing by the common factor 2 gives C_3H_7. C_6H_{14} is the molecular formula and C_nH_{2n+2} is the general formula.

QUICK CHECK

What is the correct name of $(CH_3)_2CHCH(CH_3)CH_2CH_3$?

- 2-methyl-3-methylpentane
- 3,4-dimethylpentane
- 2-ethyl-3-methylbutane
- 2,3-dimethylpentane

The longest chain has five carbons, with methyl groups on carbons 2 and 3 when numbered from the end nearer the branches. Identical groups are combined with di- and every position is given. "2-ethyl…butane" misses the longest chain.

QUICK CHECK

Which pair of compounds are functional group isomers?

- propanal and propanone
- propan-1-ol and propan-2-ol
- butane and 2-methylpropane
- ethane and ethene

Both are C_3H_6O, but one is an aldehyde and the other a ketone. The propanols are position isomers, butane and methylpropane are chain isomers, and ethane and ethene have different molecular formulae, so they are not isomers at all.

QUICK CHECK

Pentane boils at 36 °C and 2,2-dimethylpropane at 10 °C. Which statement explains the difference?

- Pentane has more electrons per molecule.
- Pentane molecules have a larger surface area of contact, so the London forces between them are stronger.
- More covalent bonds must be broken to boil pentane.
- Pentane molecules are polar but 2,2-dimethylpropane molecules are not.

They are isomers, C_5H_{12}, so they have the same number of electrons. The difference comes from shape: the compact, branched molecule touches its neighbours over a smaller area. No covalent bonds break on boiling, and both are non-polar.

QUICK CHECK

Which equation correctly shows the complete combustion of propane?

- $C_3H_8 + 3½O_2 \rightarrow 3CO + 4H_2O$
- $C_3H_8 + 4O_2 \rightarrow 3CO_2 + 4H_2$
- $C_3H_8 + 5O_2 \rightarrow 3CO_2 + 8H_2O$
- $C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$

3 C give $3CO_2$; 8 H give $4H_2O$; the right-hand side then has 6 + 4 = 10 O atoms, which is $5O_2$. The first option is incomplete combustion (and does not balance), and $8H_2O$ would need 16 hydrogen atoms.

QUICK CHECK

Which of these is a propagation step in the reaction of ethane with chlorine in UV light?

- $Cl_2 \rightarrow 2Cl\bullet$
- $Cl\bullet + C_2H_6 \rightarrow HCl + \bullet C_2H_5$
- $\bullet C_2H_5 + Cl\bullet \rightarrow C_2H_5Cl$
- $Cl\bullet + C_2H_6 \rightarrow C_2H_5Cl + H\bullet$

A propagation step has one radical on each side. The first option is initiation and the third is termination (two radicals in, none out). In the last option the chlorine radical would have to replace H directly, but the chlorine radical removes a hydrogen atom to form HCl.

QUICK CHECK

Why does a catalytic converter use a ceramic honeycomb coated with a thin layer of platinum and rhodium?

- To absorb carbon dioxide from the exhaust
- To raise the temperature of the exhaust gases
- To give a large surface area using only a small mass of expensive metal
- To filter out soot particles

The reactions happen on the surface of the metals, so a large area means fast conversion of CO, NO_x and hydrocarbons, and a thin coating keeps the cost down. The converter makes CO_2; it does not remove it.